

A PRACTICAL GUIDE FOR AI DEVELOPERS

Reexpress two

Quick Start



From cached hidden-states to Actionable Interpretability.

Train and calibrate a small Similarity-Distance-Magnitude (SDM) adaptor from precomputed embeddings. Score evaluation data, compare the results, and introspect the evidence for individual documents.

This instruction-following example checks whether an AI response correctly and faithfully answers the user's instructions. Here, **Class0** means **Not Verified** and **Class1** means **Verified**.

THE WORKFLOW

01 Prepare / pages 2–3

Download the prepared data, create a project, and import training and calibration splits.

02 Train / pages 4–5

Run two training iterations and identify the model's selected checkpoint and splits.

03 Score / page 6

Import two evaluation sets and score compatible datasets with the active model.

04 Introspect / pages 7–12

Compare selections, examine the calibration regions, and introspect a document's training exemplars.

Train on your Mac. This walkthrough uses the app's in-app, on-device training capability. To start from a model trained with the `reexpress_sdm` Python package, see the optional Python-model shortcut on page 14.

Reexpress two requires macOS 27 on an Apple silicon Mac. The tutorial includes precomputed features; you do not need to install Gemma or download its model weights.

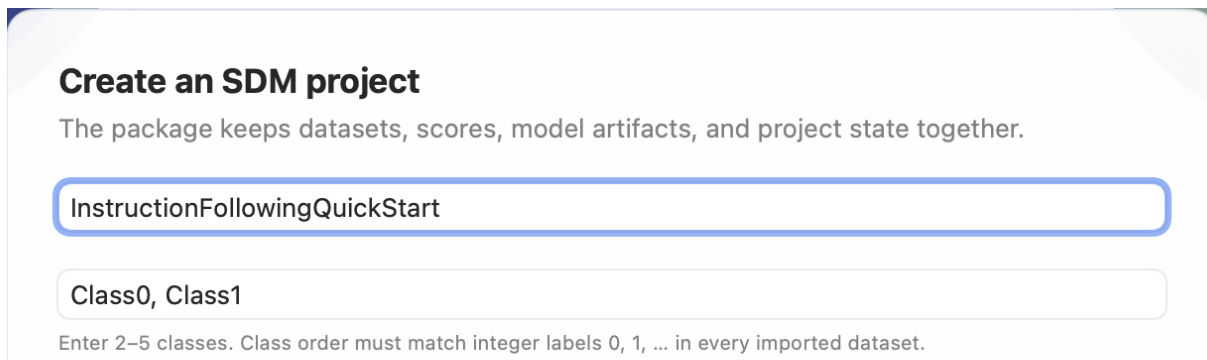
01 / PREPARE

Create a tutorial project

Download the prepared files and create a project for this walkthrough.

Open the ready-to-use sample Open the accompanying `InstructionFollowingQuickStart.sdmproject` after unzipping the downloadable `InstructionFollowingQuickStart.sdmproject.zip` and go to page 7 to inspect the same native model without training. The steps below build that project from the prepared files.

- 1 **Get the instruction-following data.**
Download and unzip the instruction-following tutorial dataset. The ZIP is about 644 MB and expands to about 1.8 GB. Allow additional space for your project, scores, and models. *(The data preparation script details the provenance of this small `OpenVerification1` sample and how the features were constructed.)*
- 2 **Create the project.**
On the welcome screen, choose **Create a project**. Enter **InstructionFollowingQuickStart**. Leave **Class names (comma-separated)** as **Class0, Class1**, in that order. **File > New Project...** opens the same flow.
- 3 **Choose its location.**
Choose **Choose package location...**, select a folder on a local internal drive outside cloud-synced folders, then **Create**. The `.sdmproject` package keeps data, models, training history, and scores together.



Create an SDM project

The package keeps datasets, scores, model artifacts, and project state together.

InstructionFollowingQuickStart

Class0, Class1

Enter 2-5 classes. Class order must match integer labels 0, 1, ... in every imported dataset.

Set the project name and keep the sample's two class names in the correct order.

Keep these four files together

Import name	JSONL file in the tutorial download
Training	<code>train.jsonl</code>
Calibration	<code>calibration.jsonl</code>
Evaluation	<code>eval.jsonl</code>
Shuffled evaluation	<code>eval_ood.jsonl</code>

01 / PREPARE

Import the training and calibration sets

Keep the two source splits separate and use the same feature settings.

Import dataset ⓘ JSON lines format

train.jsonl

Training

Split Training ↕

☐ Overwrite matching id's

IDs already present anywhere in this project are skipped. A summary reports how many were skipped.

Fields to use

- ☒ Keep document text
- ☒ Use cached embedding vectors when present
- ☐ Use attributes vectors when present

IDs, labels, and metadata are always retained. Rows missing selected features remain available for browsing, review, and export; training and scoring require complete feature vectors.

embedding_v1

Use the same fingerprint for datasets sharing a provider/model revision and attribute schema.

Cancel Import

The Training import retains document text and uses the prepared embedding vectors.

1 Import Training.

In **Data**, choose **Import Data** and open the file beginning with `train`. Name it **Training**, set **Split** to **Training**, and leave **Overwrite matching id's** off.

2 Choose the feature fields.

Leave **Keep document text** and **Use cached embedding vectors when present** selected. Deselect **Use attributes vectors when present**. Keep the fingerprint `embedding_v1`, then choose **Import**.

3 Repeat for Calibration.

Import the file beginning with `calibration`, setting name and split both to **Calibration**. Use the same feature selections and fingerprint. The tutorial data contains 4,000 Training rows and 4,000 Calibration rows.

Check before training. Both tutorial datasets should have complete 16,128-dimensional embedding features. Missing selected vectors remain pending and cannot be used for training or scoring.

02 / TRAIN

Train on your Mac

Train for two iterations of 30 epochs for this first walkthrough.

Train SDM activation

SPLITS

Training Training

Calibration Calibration

OPTIMIZATION

Presets

Start from Fresh weights

Shuffle training and calibration pool

Training iterations (J) 2

Epochs 30

Cross-entropy-equivalent epochs 1

Batch size 64

Learning rate 0.000001

Exemplar dimension 1,000

Maximum neighbors (k) 2,048

Alpha resolution 0.01

Seed 0

Training compute CPU · Accelerate

Global sample normalization, exact squared-L2 matching, lowest balanced calibration SDM loss checkpoint, canonical Float32 export.

Settings for this sample run. Scroll below this area for the **Train and calibrate** button.

1

Select the source datasets.

Open **Train**. Choose **Training** and **Calibration**, start from **Fresh weights**, and leave **Shuffle training and calibration pool** on.

2

Set the short run.

Set **Training iterations (J)** to **2** and **Epochs** to **30**. Leave the other defaults as shown, including one cross-entropy-equivalent epoch and **CPU · Accelerate**.

3

Start training.

Scroll below the settings and click **Train and calibrate**. Follow the elapsed time and completed epochs in the training monitor.

4

Let the run finish.

This example uses the completed run. Optionally, after a complete epoch, **Stop** finishes the current epoch and saves the best completed checkpoint. **Cancel** discards this training attempt.

One observed run. Training took **5 minutes and 40 seconds** on an M1 Max Mac with 64 GB memory, using CPU/Accelerate. Download, import, and evaluation scoring are separate. Runtime varies with data, hardware, and settings. *(Note: This is an abbreviated run on a small dataset for illustrative purposes. We recommend additional training for production settings.)*

02 / TRAIN

Use the selected model and splits

The winning checkpoint determines the model and its training support.

Training monitor

Saved training run

Model created:	Sep 24, 2026 at 7:37 PM
Training backend:	swift:accelerate
Total training time:	5 min 40 s
Selected iteration:	J2 of 2
Selected epoch:	29
Selected calibration balanced SDM loss:	0.14068
	Model selection criterion
Selected training balanced accuracy:	93.01%
Selected calibration balanced accuracy:	93.45%

Completed through J2, epoch 30. The curves below retain every iteration, including iterations that were not selected.

The completed native run. Its selected checkpoint is iteration 2, epoch 29.

1

Read the saved run summary.

Check **Selected iteration**, **Selected epoch**, and **Selected calibration balanced SDM loss**. The app selects the lowest class-balanced calibration SDM loss across the completed checkpoints and iterations.

2

Use the model's best iteration views.

By default, shuffling pools the imported Training and Calibration data and creates a new split for each training iteration. In Analysis, the final chosen splits are grouped under the label **Model's best iteration**. The original splits remain under **Imported datasets**.

3

Keep this model active.

Continue scoring and analysis with this model. Activating a different entry in **Model history** clears saved scores and restores that model's available best-iteration splits of the training and calibration sets; score again before comparing.

This tutorial run.

Selected iteration **2**; selected epoch **29**; highest fitted accuracy region **0.99**. These values identify the native run used here. The model separately trained as part of the tutorial for the Python package `reexpress_sdm` has its own fitted regions and results.

03 / SCORE

Score the evaluation data

Import both evaluation sets with the same embedding-only configuration.

- 1 Import Evaluation.**
In **Data**, import `eval.jsonl`. Name it **Evaluation** and choose the **Evaluation** split.
- 2 Import Shuffled evaluation.**
Import `eval_ood.jsonl`. Name it **Shuffled evaluation** and also use the **Evaluation** split. Retain document text and embeddings, deselect attributes, and use `embedding_v1` for both files. Each evaluation file contains 300 rows.
- 3 Score compatible datasets.**
Choose **Score All Datasets**. This includes compatible imported datasets and the active model's best-iteration views. After completion, check the dataset cards and any scoring details for skipped or unscored data.

Data and model artifacts ⓘ

Score All Datasets

Review changes (0)

Score All complete: 6 of 6 eligible datasets committed; 0 skipped.

Attach model sources

Optionally attach training or calibration sources for text, winning-split views, and continued training. Either source can be added later.

TRAINING SOURCES & MODEL VIEWS	CALIBRATION SOURCES & MODEL VIEWS	EVALUATION
Training ⓘ 4,000 rows · 4,000 labeled 4,000 feature-ready · 16,128D (embedding 16,128 + attributes 0)	Calibration ⓘ 4,000 rows · 4,000 labeled 4,000 feature-ready · 16,128D (embedding 16,128 + attributes 0)	Evaluation ⓘ 300 rows · 300 labeled 300 feature-ready · 16,128D (embedding 16,128 + attributes 0)
Best iteration training ⓘ Best-iteration pointer view Edits affect the original documents. The trained model retains its frozen support snapshot. 4,000 rows · 4,000 labeled 4,000 feature-ready · 16,128D (embedding 16,128 + attributes 0)	Best iteration calibration ⓘ Best-iteration pointer view Edits affect the original documents. The trained model retains its frozen support snapshot. 4,000 rows · 4,000 labeled 4,000 feature-ready · 16,128D (embedding 16,128 + attributes 0)	Shuffled evaluation ⓘ 300 rows · 300 labeled 300 feature-ready · 16,128D (embedding 16,128 + attributes 0)

Six compatible datasets are scored: four imports and two best-iteration views.

To score one dataset, use its **Score** or **Rescore** action. A row needs complete features compatible with the model's dimension and representation fingerprint. Use this same active native model throughout the comparisons below.

Start the analysis

Start with the winning calibration split as a reference.

1

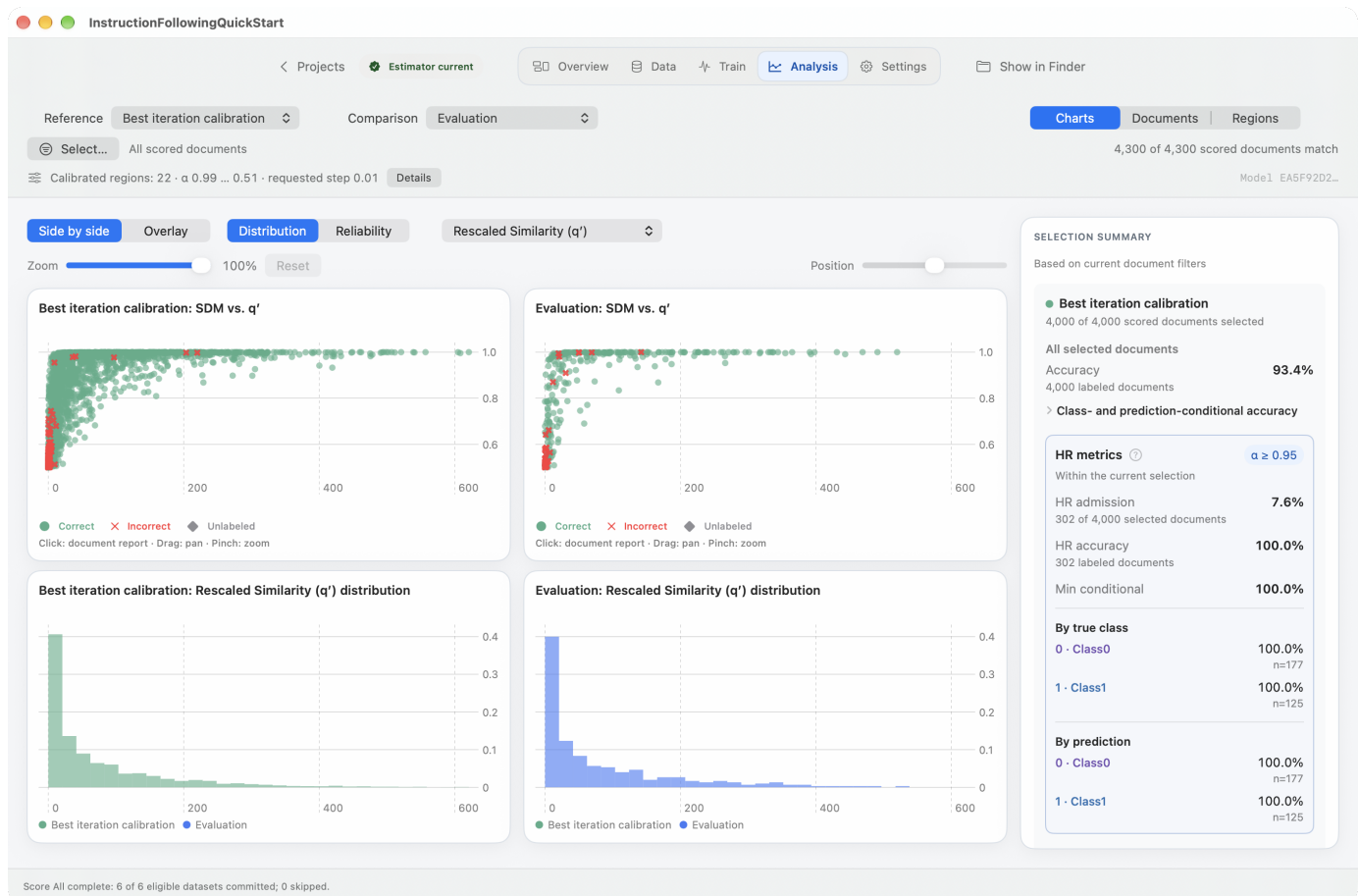
Choose datasets and clear filters.

Open **Analysis**. For **Reference**, choose **Best iteration calibration** under **Model's best iteration**. For **Comparison**, choose **Evaluation** under **Imported datasets**. Select **Charts**. *The winning calibration split is used to partition the data into class- and prediction-conditional HR ("High-Reliability") regions, so it is the typical first place to start the analysis.*

2

Read the summary in the right panel.

All selected documents reports accuracy for the selected labeled rows. Click the **Class- and prediction-conditional accuracy** disclosure for the per-class values and cell counts, as well as the **Min conditional** value. These summary statistics reflect the documents that match the given selection, which in this case is all of the documents since all documents are currently selected.



The Charts tab compares selected datasets. The right-hand summary provides aggregate statistics for all selected documents, as well as for those from the HR subset at the currently selected α value.

Understanding the selection mechanism

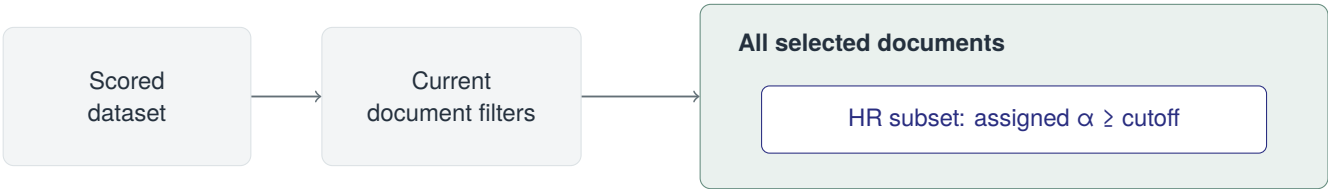
Each scored document has a prediction and an assignment to a region of the distribution estimated to have a class- and prediction-conditional accuracy at least a given α . The **Analysis** selection mechanism enables filtering the documents by various criteria.

Three key quantities

Quantity	How to read it
Predicted-class SDM	The SDM activation output for the predicted class, shown on the scatter plot.
Assigned region	The document’s calibrated class- and prediction-conditional accuracy region, using either Centroid or DKW lower . No region indicates no calibrated region > 0.5 was assigned. Such a document will still have a prediction, but it is out-of-distribution to the SDM estimator.
HR metrics threshold	The currently active cutoff for α used to summarize admission and labeled accuracy within the current selection.

Understanding the α level. An assigned α describes a calibrated class- and prediction-conditional accuracy region relative to the **Best iteration calibration** split. For example, a value of 0.99 is an estimate that the datapoint falls into a region of the distribution for which the class- and prediction-conditional accuracy is at least 0.99. It does not guarantee that an individual prediction is correct, nor does it guarantee robustness against adversarial inputs.

Filtering documents changes the denominator



Filtering documents alters the denominator of the displayed summary statistics.

Admission is the proportion of evaluated documents whose assigned region meets or exceeds the selected α threshold. Here, evaluated documents means every scored document in the *current selection*, including unlabeled rows.

What is “conditional” accuracy? Class-conditional accuracy groups documents by their known label. Prediction-conditional accuracy groups them by the model’s prediction. The displayed values only consider documents with known labels.

The α threshold in the right panel is sticky. The **Selection Summary** panel’s α used for summary statistics always uses the most recent **Minimum accepted α** from the **Select documents** dialog box. Clicking **Clear filters** resets the graphs to include all documents, but retains the most recent threshold for the **HR metrics** subset in the **Selection Summary** panel on the right side of the window.

04 / INTROSPECT

Compare the two evaluation sets

Hold the model and selection rules fixed while changing the dataset.

1

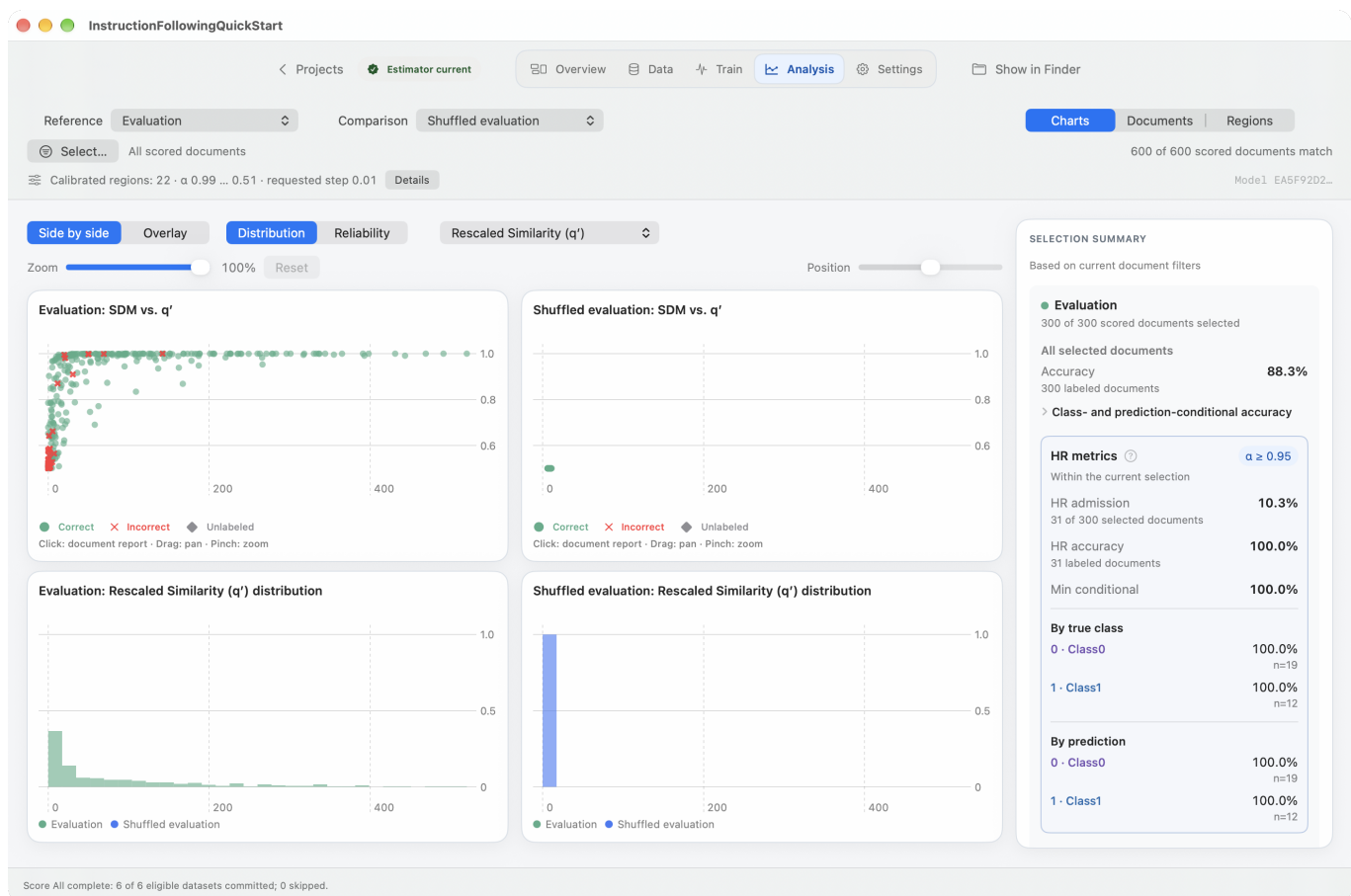
Set the comparison.

Choose **Evaluation** as **Reference** and **Shuffled evaluation** as **Comparison**. Both are under **Imported datasets**. Choose **Clear filters**, if applicable.

2

Set a cutoff, then restore the full selection.

Open **Select...**. Set **HR subset** to **Minimum accepted α** ; choose **0.95** from **Fitted α** . Switch back to **All HR levels**, retaining that cutoff for HR metrics. Keep **Estimator** at **Centroid**.



At centroid $\alpha \geq 0.95$, the Evaluation set admits 31 of 300 documents (10.3%). The Shuffled evaluation set admits none (0%). These are results from this sample run.

Compare admission and feature distributions. The characters of the model summary and explanation in the shuffled set have been randomly permuted to serve as an out-of-distribution stress case. Its ground-truth labels are all Class0, so some conditional cells can be undefined.

Introspect one prediction

Importantly, the analysis is not limited to aggregate statistics. Instance-wise signals can be examined for each document.

1

Open a result.

Click a chart point or select a row in **Documents**. Choose **Report** for the document report. Alternatively, **Show row** jumps to the selected document's row within the table view given the current filters and sort.

Reexpress two Document report 1 of 1 on this page Done

CLASSIFICATION RESULT

Prediction
Class1 (1)

Provided label (when scored)
Class0 (0)

Label agreement (when scored)
Incorrect

Class- and prediction-conditional accuracy region (centroid)
 $\alpha \geq 0.75$

Accuracy region accounting for distance-eCDF uncertainty (DKW lower)
 $\alpha \geq 0.75$

An assigned α describes the estimator's calibrated accuracy region, not certainty about this individual document. Shifted data can still be miscalibrated.

DOCUMENT A 16 pt A

<topic> The user asked to evaluate the arithmetic expression $2 \cdot 7 + 5 \cdot 11 + 9 \cdot 2 + 7 \cdot 4$. The AI response computed each product and summed them to get 117. </topic> Does the following model explanation agree that the response is correct? <model_explanation> The arithmetic is straightforward and correct: $14 + 55 + 18 + 28 = 117$. The response directly answers the question. </model_explanation> Answer with a single word: Yes or No?

A label disagreement to inspect. The report shows the provided label and this model's prediction.

2

Read the prediction in context.

Compare the prediction with the provided label, then examine the assigned region and the retained document text. Here, the retained text is the agreement prompt used to generate the embeddings, including a summary and explanation of the original response. Scroll farther down for the instance-level signals that determined the uncertainty estimate.

Introspect the training support

Review the nearest training exemplars alongside the query document.

1

Open the support section.

In the document report, scroll to **Nearest training exemplars**. These exact matches come from the winning model's saved training support, which may differ from the original Training set import.

Reexpress two Document report < 1 of 1 on this page > Done

NEAREST TRAINING EXEMPLARS

Exact matches against the active model's saved training support, after its selected shuffle—not necessarily the original Training dataset. Labels and predictions below are the model's training snapshot. A red Training error badge marks an exemplar whose training prediction was incorrect, not an uncertainty region.

#1 pos_1085751_07ccbae0-c8b0-43cb-8abc-082db1812ff1 d, 5.044

Label: Class1 (1) · model prediction: Class1 (1)

<topic> The question asks to solve for x given $x/y = 10/4$ and $y = 18$. The AI response substitutes $y = 18$, computes $x = (10/4) \cdot 18 = 45$, and gives 45 as the final answer. </topic>
 Does the following model explanation agree that the response is correct?
 <model_explanation> The algebra is straightforward and correct: $x/18 = 10/4$ implies $x = 18 \dots$

[Show full text](#)

#2 pos_547135_f8008495-6151-4c29-aff0-56e39ef1615d d, 5.675

Label: Class1 (1) · model prediction: Class1 (1)

<topic> The user asked to calculate 2015 divided by $(2015^2 - 2016 \times 2014)$. The AI response used the difference of squares to simplify the denominator to 1 and concluded the value is 2015. </topic> Does the following model explanation agree that the response is correct? <model_explanation> The algebra is straightforward and correct: $2016 \times 2014 = (2 \dots$

[Show full text](#)

The first two nearest training exemplars for the same inspected document and active native model. Choose **Show full text** to expand an exemplar, or scroll to inspect additional matches.

2

Compare text, labels, and predictions.

Look for useful similarities and differences in the retained text. Check whether the support labels and predictions agree, and use that context to decide what warrants further review.

Use exemplars as context. They help you examine the model's support, but they should be interpreted with the signals of Similarity, Distance, and Magnitude, and the estimated HR region. As the epistemic (reducible) uncertainty increases, the exemplars will tend to have a looser connection to the query document. *Note that support documents exclude their self-match when scored and matched. I.e., the list of nearest exemplars for a document from **Best iteration training** does not include a match to itself.*

Examine exact and cumulative regions

Use Regions to examine individual accuracy regions.

- 1
- Start with the full selection.**

Choose **Regions** and clear document filters. The **All selected documents** row provides a summary that includes rejected and unlabeled rows in its count.

Per region (exact α)Cumulative ($\alpha \geq x$)?

0: Class0 · 1: Class1

Evaluation									
Region	n	% of selection	Accuracy	y = 0	y = 1	$\hat{y} = 0$	$\hat{y} = 1$	Min conditional	
All selected documents	300	—	88.3%	92.3% n=143	84.7% n=157	84.6% n=156	92.4% n=144	84.6%	
0.99	2	0.7%	100.0%	100.0% n=2	— n=0	100.0% n=2	— n=0	—	Documents
0.98	0	0.0%	—	— n=0	— n=0	— n=0	— n=0	—	Documents
0.96	2	0.7%	100.0%	100.0% n=2	— n=0	100.0% n=2	— n=0	—	Documents
0.95	27	9.0%	100.0%	100.0% n=15	100.0% n=12	100.0% n=15	100.0% n=12	100.0%	Documents

Top of the exact region table. The full-selection summary includes all 300 Evaluation documents, with 88.3% accuracy.

View	Population and denominator
All selected documents	Every scored row matching the current selection. No α cutoff applies, so the Admission / % of selection cell shows a dash.
Per region (exact α)	Rows assigned exactly that region. % of selection divides its count by the full current selection. The none row contains rejected documents.
Cumulative ($\alpha \geq x$)	Rows assigned to regions at or above the cutoff. Admission divides their count by the full current selection.

- 2
- Move from a region’s summary to its documents.**

Choose **Documents** on an exact region row, or **Documents $\geq \alpha$** on a cumulative row. This switches to a Documents table with that region or cutoff applied.
- 3
- Check the class cells.**

The cells show conditional accuracy and labeled counts. An absent class has undefined accuracy, rather than zero. Unlabeled rows contribute to counts and admission, but not accuracy. These statistics use every applicable document within a filtered subset, whereas the scatter plots show at most 2,000 points.

OPTIONAL / UPDATABILITY

Modify a document's label or dataset

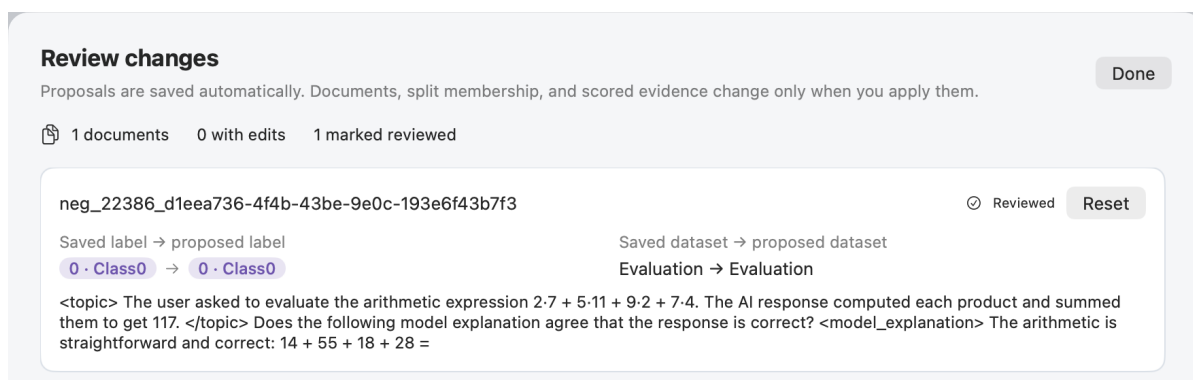
The app uses a queuing system to stage updates (label or dataset changes), since some changes trigger a rescoring or retraining.

1 Choose a document to review.

Use **Clear filters**, then in **Select...** try **Incorrect only** or **Possible label errors**. Inspect the document and support. The latter is a candidate-finding heuristic; a label/prediction disagreement is not proof that the label is wrong.

2 Stage a practice item.

For your chosen document, select the checkmark **Reviewed in this batch**, leaving **Proposed label** and **Proposed destination** unchanged. For a real edit, set those fields or use **Stage edits...** (**Analysis > Documents**) for multiple documents. Proposals save immediately.



A reviewed-only practice item. The saved and proposed label and dataset remain unchanged.

3 Inspect the pending batch.

Open **Review changes** (upper-right of the window). Check the saved and proposed labels and destinations. Use **Reset** for an individual proposal or **Discard all...** for the batch. Closing the sheet leaves proposals saved.

4 Finish the practice.

For this walkthrough, choose **Discard all...** and confirm, preserving the tutorial baseline. In a real review, choose **Apply changes...** and confirm the listed documents when ready. Resolve conflicts before applying.

Some changes necessitate score or model refreshes. Reviewed-only updates preserve scores; the **Reviewed in this batch** flag can be used as a temporary marker when reviewing documents. In contrast, a label or dataset change invalidates affected scores once applied, requiring rescoring. Edits to training or calibration data can make the model stale, requiring retraining. The app provides estimator status (see the top navigation bar), indicating when a retraining is required.

Saving, reopening, and...next steps

Keep the project as your working record and export the artifacts you need. The model and scores are interoperable with the Python package `reexpress_sdm`.

Reopen the complete project

Choose **Data > Show Project in Finder** to locate `InstructionFollowingQuickStart.sdmproject`. After closing the app, reopen it with **File > Open Project...** or by opening the package in Finder. *(Note that only one project can be open at a time. Quit the app before copying an `.sdmproject` file as a backup or opening a different copy.)*

Export models and scored data

In **Data**, use the model export menu to save either a model with its support content or a content-redacted version. An `.sdmkitmodel` includes the model; it is different from the complete project that also includes the evaluation datasets. *(The Python package `reexpress_sdm` uses `.sdmkitmodel` model files.)*

Use a dataset card's **Export dataset** menu to save the source data, or **Export scored dataset · Python / app** to save source features together with saved score diagnostics. Both offer JSONL and `.sdmdataset` bundles. **Export matching...** in Analysis exports every matching document in the current selection, not just the visible page.

Before sharing. A full model may include retained training/support text and identifiers. Content redaction removes selected content; it does not make embeddings or model data anonymous.

Optional shortcut: Start from a Python-trained model

To explore a model trained with the Python package `reexpress_sdm` (or to otherwise import an `.sdmkitmodel` exported from this app), choose **File > Start Project from Model...** and select the supplied `.sdmkitmodel`.

Once the model is imported, proceed to import compatible evaluation data and score it as above. Original training/calibration source files are optional for scoring new documents. Use **Attach model sources** when you need the original source text or features for inspection, or to retrain or continue training.

Guides and research

- [Reexpress guides and tutorial resources](#). Also available from **Help > Documentation at re.express**.
- [Python getting-started tutorial](#). Open the link to view the rendered tutorial.
- [Similarity-Distance-Magnitude Activations](#).
- [Research Note: Nested Similarity-Distance-Magnitude Estimators](#).

Reexpress two 0.1.4. This guide's observed native run used CPU/Accelerate on macOS 27. It is a worked example, not a performance guarantee for other data or hardware.